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Chapter 6: Extra proofs and comments

NOTE:

1. These notes do not contain any new examinable material.
2. They contain some elaborations on the lecture notes, with extra explanations and examples, and even a few mathematical derivations—which are not examinable.
3. I will refer to them in class.
4. However, students can ignore them if they don't find them useful.

Slide 6-4

Imports of goods & services (% of GDP)

Country	% of GDP
United States	~14.3% worldbank
Japan	~23.6% worldbank
Canada	~32.7% worldbank
France	~34.2% worldbank
Germany	~37.7% worldbank

(All values are the most recent available, mainly 2024 estimates from the World Bank dataset.)

Slides 6-17 and 6-18

Slide 6-17 gives notation, slide 6-18 gives prices.

Slide 6-17	Slide 6-18	Price
USD/EUR	1.1815	1.1815 USD per EUR
JPY/USD	106.25	106.25 JPY per USD
USD/GBP	1.3188	1.3188 USD per GBP
USD/AUD	0.7222	0.7222 USD per AUD

The price is in terms of the currency given in the numerator, and it's always “per one” unit of the denominator.

You don't have to remember whether a quote uses the “American term” or the “European term”.

Slide 6-21

NOTE: when we write, e.g., 1 GBP = 1.3183 USD, the GBP is the *base currency*, and the USD is the *quote currency*.

So, if there's a 1 in front of the number, it's the base currency; the term with the other number (the price) is the quote currency.

Slide 6-27

Let's first consider cross currencies *without* the bid-ask spread.

Suppose we have two exchange rates:

BRL 0.9/USD and THB 25/USD; i.e.,

1 USD = 0.9 BRL, and 1 USD = 25 THB.

Let's use 0.9 BRL to buy THBs.

0.9 BRL = 1 USD = 25 THB.

So, 0.9 BRL = 25 THB, *or*

$$1 \text{ BRL} = \frac{25}{0.9} = 27.78 \text{ THB.}$$

Slide 6-27

Just to clarify,

For the Brazilian real,

- the bid rate is BRL 0.9955/USD, (i.e., $1 \text{ USD} = 0.9955 \text{ BRL}$)
- the ask rate is BRL 1.0076/USD, (i.e., $1 \text{ USD} = 1.0076 \text{ BRL}$);

for the Thai baht,

- the bid rate is THB 25.2513/USD, (i.e., $1 \text{ USD} = 25.2513 \text{ THB}$)
- the ask rate is THB 25.3986/USD, (i.e., $1 \text{ USD} = 25.3986 \text{ THB}$).

Slide 6-28

Suppose you want to use Brazilian reais to buy Thai bahts.

First, use reais to buy US dollars, then use US dollars to buy bahts.

To buy USD, do we pay 0.9955 BRL or 1.0076 BRL?

Once we have USD, we sell them for Thai bahts.

To sell USD, do we receive 25.2513 THB or 25.3986 THB?

Conclusion 1:

When we use BRL to buy THBs, we find that

$$1.0076 \text{ BRL} = 1 \text{ USD} = 25.2513 \text{ THB};$$

$$1 \text{ BRL} = \frac{25.2513}{1.0076} = \frac{\text{bid rate for Thai baht / USD}}{\text{ask rate for Brazilian real / USD}} = 25.0608 \text{ THB}.$$

Now, let's use Thai bahts to buy Brazilian reais.

First, use bahts to buy US dollars, then use US dollars to buy reais.

To buy USD, do we pay 25.2513 THB or 25.3986 THB?

Once we have USD, we sell them for Brazilian reais.

To sell USD, do we receive 0.9955 BRL or 1.0076 BRL?

Conclusion 2:

When we use THBs to buy BRLs, we find that

$$25.3986 \text{ THB} = 1 \text{ USD} = 0.9955 \text{ BRL};$$

$$1 \text{ BRL} = \frac{25.3986}{0.9955} = \frac{\text{ask rate for Thai baht / USD}}{\text{bid rate for Brazilian real / USD}} = 25.5134 \text{ THB}.$$

Slide 6-30

Bid price for GBP in New York is USD 1.9422

(i.e., sell 1 GBP $\leftrightarrow$ buy 1.9422 USD)

Ask price for EUR in Frankfurt is USD 1.4925

(i.e., buy 1 EUR $\leftrightarrow$ sell 1.4925 USD)

Ask price for GBP in London is EUR 1.2998

(buy 1 GBP $\leftrightarrow$ sell 1.2998 EUR).

Is there an arbitrage opportunity?

Let's start by *selling* USD and see if we can finish by *buying* USD.

(in Frankfurt) sell 1,000,000 USD (ask price)

$$\text{buy } \frac{1}{1.4925} \times 1,000,000 \text{ EUR}$$

(in London) sell $\frac{1}{1.4925} \times 1,000,000 \text{ EUR}$ (ask price)

$$\text{buy } \frac{1}{1.4925 \times 1.2998} \times 1,000,000 \text{ GBP}$$

(in New York) sell $\frac{1}{1.4925 \times 1.2998} \times 1,000,000 \text{ GBP}$ (bid price)

$$\text{buy } 1.9422 \times \frac{1}{1.4925 \times 1.2998} \times 1,000,000 \text{ USD} = 1,001,159.05 \text{ USD.}$$

Slide 33

More on forward contracts.

The *forward price* is chosen so that the forward contract has zero value today, i.e., the forward price is set (by market supply and demand, or by arbitrage) so that no money need change hands today. Money is only paid on the delivery date.

The one who agrees to *buy* the underlying asset (GBPs, in our example) at the future date has a *long* position; the one who agrees to *sell* the asset at the future date has a *short* position.

(Mnemonic device: “long good buy” versus *short sell*.)

Suppose the forward contract matures at date T .

Write e_T for the “spot price” (i.e., the actual price of GBPs) on the delivery date.

Write f_0 for the delivery (forward) price that was locked in at date 0.

In our example, we are *short* pounds, so we hedge this by getting a *long* forward contract.

Suppose we take a *long* position in this forward contract for N units of the foreign currency.

At date T , pay Nf_0 USD for the foreign currency, and then sell them on the market, receiving Ne_T USD.

Total payoff from *long* forward contract: $N(e_T - f_0)$.

(If you graph this as a function of e_T , it is a straight line with positive slope.)

Now, if you also must *pay* N units of the foreign currency, which costs Ne_T (at the current market price), then

$$\text{Total cash-flow } N(e_T - f_0) - Ne_T = -Nf_0 .$$

So, you've locked in a payment of Nf_0 .

(Total payoff from *short* forward contract is $N(f_0 - e_T)$, which gives a straight line with negative slope.)